

[illegible]

```

Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.insertMany([
  {
    title: "Atomic Habits",
    author: "James Clear",
    published_year: 2018,
    format: "Paperback",
    technical_specs: {
      pages: 320,
      language: "English",
      weight: "450g"
    },
    member_id: 102,
    reviews: [
      {
        member_id: 102,
        rating: 5,
        comments: "Great book for building habits."
      },
      {
        member_id: 104,
        rating: 5,
        comments: "Very useful and motivating."
      }
    ]
  },
  {
    title: "Project Hail Mary",
    author: "Andy Weir",
    published_year: 2021,
    format: "Digital",
    technical_specs: {
      pages: 496,
      language: "English",
      file_size: "8 MB"
    },
    member_id: 103,
    reviews: [
      {
        member_id: 103,
        rating: 5,
        comments: "Amazing science fiction story."
      },
      {
        member_id: 105,
        rating: 4,
        comments: "Excellent storytelling."
      }
    ]
  },
  {
    title: "The Midnight Library",
    author: "Matt Haig",
    published_year: 2020,

```

```

    {
      title: "The Midnight Library",
      author: "Matt Haig",
      published_year: 2020,
      format: "Hardcover",
      technical_specs: {
        pages: 304,
        language: "English",
        weight: "500g"
      },
      member_id: 104,
      reviews: [
        {
          member_id: 104,
          rating: 4,
          comments: "Interesting concept."
        },
        {
          member_id: 106,
          rating: 3,
          comments: "Good but somewhat slow."
        }
      ]
    },
  ],
  {
    title: "Klara and the Sun",
    author: "Kazuo Ishiguro",
    published_year: 2021,
    format: "Digital",
    technical_specs: {
      pages: 320,
      language: "English",
      file_size: "6 MB"
    },
    member_id: 105,
    reviews: [
      {
        member_id: 105,
        rating: 4,
        comments: "Beautiful and thoughtful."
      },
      {
        member_id: 107,
        rating: 5,
        comments: "A beautifully written story."
      }
    ]
  }
]
}
}

class atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.find({
  "reviews.rating": { $gt: 4 }
})

{
  _id: ObjectId('6a7ad9bc527fa4c955249b89'),
  title: 'The Digital Future',

```

```

Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.find({
  |   "reviews.rating": { $gt: 4 }
  | })
[
  {
    _id: ObjectId('6a7ad9bc527fa4c955249b09'),
    title: 'The Digital Future',
    published_year: 2022,
    formats: [ 'Digital', 'Paperback' ],
    member_id: 101,
    reviews: [
      { rating: 5, comment: 'Excellent book' },
      { rating: 4, comment: 'Very useful' }
    ]
  },
  {
    _id: ObjectId('6a7ad9c8527fa4c955249b0b'),
    title: 'Modern Web Development',
    published_year: 2023,
    formats: [ 'Digital', 'Hardcover' ],
    member_id: 103,
    reviews: [
      { rating: 5, comment: 'Excellent web development guide' },
      { rating: 5, comment: 'Very informative' }
    ]
  },
  {
    _id: ObjectId('6a83d4b8c4b0b60db5874971'),
    title: 'The Psychology of Money',
    author: 'Morgan Housel',
    published_year: 2020,
    format: 'Digital',
    technical_specs: { pages: 256, language: 'English', file_size: '5 MB' },
    member_id: 101,
    reviews: [
      {
        member_id: 101,
        rating: 5,
        comments: 'Excellent book about money and behavior.'
      },
      {
        member_id: 102,
        rating: 4,
        comments: 'Very practical and easy to understand.'
      }
    ]
  },
  {
    _id: ObjectId('6a83d532c4b0b60db5874972'),
    title: 'Atomic Habits',
    author: 'James Clear',
    published_year: 2018,
    format: 'Paperback',
    technical_specs: { pages: 320, language: 'English', weight: '450g' },
    member_id: 102,
    reviews: [

```

```

    member_id: 102,
    reviews: [
      {
        member_id: 102,
        rating: 5,
        comments: 'Great book for building habits.'
      },
      { member_id: 104, rating: 5, comments: 'Very useful and motivating.' }
    ]
  },
  {
    _id: ObjectId('6a83d532c4b0b60db5874973'),
    title: 'Project Hail Mary',
    author: 'Andy Weir',
    published_year: 2021,
    format: 'Digital',
    technical_specs: { pages: 496, language: 'English', file_size: '8 MB' },
    member_id: 103,
    reviews: [
      {
        member_id: 103,
        rating: 5,
        comments: 'Amazing science fiction story.'
      },
      { member_id: 105, rating: 4, comments: 'Excellent storytelling.' }
    ]
  },
  {
    _id: ObjectId('6a83d532c4b0b60db5874975'),
    title: 'Klara and the Sun',
    author: 'Kazuo Ishiguro',
    published_year: 2021,
    format: 'Digital',
    technical_specs: { pages: 320, language: 'English', file_size: '6 MB' },
    member_id: 105,
    reviews: [
      { member_id: 105, rating: 4, comments: 'Beautiful and thoughtful.' },
      { member_id: 107, rating: 5, comments: 'A beautifully written story.' }
    ]
  }
]
Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.find({
  |   "reviews.rating": { $gt: 4 }
  | }, {
  |   title: 1,
  |   "reviews.rating": 1,
  |   _id: 0
  | }).pretty()
[
  {
    title: 'The Digital Future',
    reviews: [ { rating: 5 }, { rating: 4 } ]
  },
  {
    title: 'Modern Web Development',
    reviews: [ { rating: 5 }, { rating: 5 } ]
  }
]

```

```

    {
      _id: ObjectId('6a83d532c4b0b60db5874972'),
      title: 'Atomic Habits',
      author: 'James Clear',
      published_year: 2018,
      format: 'Paperback',
      technical_specs: { pages: 320, language: 'English', weight: '450g' },
      member_id: 102,
      reviews: [
        {
          member_id: 102,
          rating: 5,
          comments: 'Great book for building habits.'
        },
        { member_id: 104, rating: 5, comments: 'Very useful and motivating.' }
      ]
    },
    {
      _id: ObjectId('6a83d532c4b0b60db5874973'),
      title: 'Project Hail Mary',
      author: 'Andy Weir',
      published_year: 2021,
      format: 'Digital',
      technical_specs: { pages: 496, language: 'English', file_size: '8 MB' },
      member_id: 103,
      reviews: [
        {
          member_id: 103,
          rating: 5,
          comments: 'Amazing science fiction story.'
        },
        { member_id: 105, rating: 4, comments: 'Excellent storytelling.' }
      ]
    },
    {
      _id: ObjectId('6a83d532c4b0b60db5874975'),
      title: 'Klara and the Sun',
      author: 'Kazuo Ishiguro',
      published_year: 2021,
      format: 'Digital',
      technical_specs: { pages: 320, language: 'English', file_size: '6 MB' },
      member_id: 105,
      reviews: [
        { member_id: 105, rating: 4, comments: 'Beautiful and thoughtful.' },
        { member_id: 107, rating: 5, comments: 'A beautifully written story.' }
      ]
    }
  ]
}
Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.find({
|   format: { $regex: "Digital" }
| })
[
  {
    _id: ObjectId('6a83d4b8c4b0b60db5874971'),
    title: 'The Psychology of Money',

```

```

Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.find([
  {
    format: { $regex: "Digital", $options: "i" }
  })
]
[
  {
    _id: ObjectId('6a83d5b8c0b66db5874971'),
    title: 'The Psychology of Money',
    author: 'Morgan Housel',
    published_year: 2020,
    format: 'Digital',
    technical_specs: { pages: 256, language: 'English', file_size: '5 MB' },
    member_id: 101,
    reviews: [
      {
        member_id: 101,
        rating: 5,
        comments: 'Excellent book about money and behavior.'
      },
      {
        member_id: 102,
        rating: 4,
        comments: 'Very practical and easy to understand.'
      }
    ]
  },
  {
    _id: ObjectId('6a83d532c0b66db5874973'),
    title: 'Project Hail Mary',
    author: 'Andy Weir',
    published_year: 2021,
    format: 'Digital',
    technical_specs: { pages: 496, language: 'English', file_size: '8 MB' },
    member_id: 103,
    reviews: [
      {
        member_id: 103,
        rating: 5,
        comments: 'Amazing science fiction story.'
      },
      { member_id: 105, rating: 4, comments: 'Excellent storytelling.' }
    ]
  },
  {
    _id: ObjectId('6a83d532c0b66db5874975'),
    title: 'Klara and the Sun',
    author: 'Kazuo Ishiguro',
    published_year: 2021,
    format: 'Digital',
    technical_specs: { pages: 320, language: 'English', file_size: '6 MB' },
    member_id: 105,
    reviews: [
      { member_id: 105, rating: 4, comments: 'Beautiful and thoughtful.' },
      { member_id: 107, rating: 5, comments: 'A beautifully written story.' }
    ]
  }
]

```

```

Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.aggregate([
  {
    $match: {
      published_year: { $gt: 2020 }
    }
  },
  {
    $unwind: "$reviews"
  },
  {
    $group: {
      _id: "$title",
      average_rating: {
        $avg: "$reviews.rating"
      }
    }
  }
])
[
  { _id: 'The Digital Future', average_rating: 4.5 },
  { _id: 'Project Hail Mary', average_rating: 4.5 },
  { _id: 'Modern Web Development', average_rating: 5 },
  { _id: 'Mlara and the Sun', average_rating: 4.5 }
]
Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.createIndex({
  "reviews.comments": "text"
})
reviews.comments_text
Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'reviews.comments_text',
    weights: { 'reviews.comments': 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  }
]
Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.find({
  $text: {
    $search: "excellent"
  }
})
[
  {
    _id: ObjectId('6a83d532c4b0b60db5874973'),
    title: 'Project Hail Mary',
    author: 'Andy Weir',
    published_year: 2021,
    format: 'Digital',
    technical_specs: { pages: 496, language: 'English', file_size: '8 MB' },
    member_id: 103,
  }
]

```

```

}
Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> db.book_metadata.find({
  $text: {
    $search: "excellent"
  }
}).explain()
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'excellent',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: 'CF6F4CEE',
    planCacheShapeHash: 'CF6F4CEE',
    planCacheKey: '08852285',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      indexPrefix: {},
      indexName: 'reviews.comments_text',
      parsedTextQuery: {
        terms: [ 'excel' ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      },
    },
    textIndexVersion: 3,
    inputStage: {
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { _fts: 'text', _ftsx: 1 },
        indexName: 'reviews.comments_text',
        isMultikey: true,
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'backward',
        indexBounds: {}
      }
    }
  },
  rejectedPlans: []
}

```

```

},
queryShapeHash: 'F0F10B238B9A2810CDAC1E21EC4982510A79717B873B97D84CC5291AD3',
command: {
  find: 'book_metadata',
  filter: { '$text': { '$search': 'excellent' } },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'ac-kyzhqe-shard-00-01.ge7tc4r.mongoddb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787026452, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('V9mEi9aWb043iveu0W0vy+PksSI=', 0),
    keyId: Long('7610901413966643201')
  }
},
operationTime: Timestamp({ t: 1787026452, i: 1 })
}
Atlas atlas-s2e6hn-shard-0 [primary] BookFlow> ␣

```